RANJANA S

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Executive Summary

ECE undergraduate with a strong interest in VLSI, Embedded Systems, IoT, and microcontrollers. Hands-on experience with digital design, FPGA, sensors, and embedded projects. Seeking an opportunity to apply technical skills and grow in a practical engineering environment.

Skills

Programming languages:

- HDL: Verilog
- C
- Java
- HTML
- CSS

EDA & Simulation Tools:

- LTSpice
- ModelSim
- QuestaSim
- Xilinx Vivado
- Intel Quartus Prime
- EDAPlayground
- DSCH3
- Microwind3

Embedded Systems & IoT:

- Arduino
- ESP32
- ESP8266
- Blynk IoT

Education

St. Joseph Girls Higher Secondary School, Salem
Class 10th | 2020-2021
Percentage : All Pass

St. Joseph Girls Higher Secondary School, Salem
Class 12th | 2022-2023
Percentage : 80.3%

Knowledge Institute of Technology
Electronics and Communication Engineering | 2023- 2027
CGPA(Upto 6th Semester): 8.48

Projects

1.Smart IoT Energy Meter | ESP32, Arduino IDE, Blynk IoT: Developed an ESP32-based IoT energy meter for real time monitoring of voltage, current, power, and energy using Blynk.

2.Energy Recovery circuit using Adiabatic Logic: Designed and simulated a low-power adiabatic CMOS energy recovery circuit using LTspice and Microwind.

3.Digital Calculator using Intel Quartus & DE10 Lite FPGA: Implemented an 8-bit Verilog-based digital calculator on the DE10-Lite FPGA with seven-segment display output.

4.Grass Cutting Robot: Developed an ESP32-based wireless grass-cutting robot with Wi-Fi control, obstacle detection, and autonomous navigation.

Internships

- **Sans Innovations, Salem – Embedded System Design in Industries:** Completed hands-on training on IoT fundamentals, sensor interfacing, and microcontroller-based data acquisition for practical applications.
- **Web Development Intern | Nibode Technologies:** Completed hands-on training in HTML, CSS, responsive web design, and basic front-end development for creating and styling web pages.
- **Government ITI, Salem – Skill Development Programme on Internet of Things (IoT):** Completed hands-on training on IoT fundamentals, sensor interfacing, and microcontroller-based data acquisition for practical applications.

Certifications & Technical Activities

1. Completed **Advanced Digital System Design** course from HCL GUVI.
2. Completed **Introduction to FPGA Based Digital System Design** course from HCL GUVI.
3. Completed **Digital Design Verification with System Verilog** course from HCL GUVI.
4. Completed **Low Power VLSI design** course from HCL GUVI.
5. Secured first place in Engineer Clinic II for presenting an ESP32-based **grass cutting robot** with wireless control and obstacle detection - 2024.
6. Presented a technical paper on an **e-commerce website** for direct market access at Paavai Engineering College-2024.
7. Presented a technical paper on a **tooth-audible device** at Muthayammal College of Engineering - 2025.
8. Presented a paper on "**Energy Recovery Circuit with Improved Diode Free Adiabatic Logic**" at the ICOST-2026 conference, Annapoorana Engineering College, Salem.